

The Propagation Framework

How Reality Derives Itself from Three Axioms

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The Claim

Everything is propagation. Matter is stable self-reinforcing wave patterns. Forces are refraction. Time is what propagation feels like from inside. Three axioms about how disturbances move through a medium derive the generation structure of the Standard Model, the Koide mass ratio, the Weinberg angle, and gravity - with zero free parameters.

Three Axioms

Axiom 1: Propagation is Fundamental

Everything that exists propagates. The medium is not empty space but a field capable of carrying a signal.

Axiom 2: Finite Causal Velocity

Every medium has a maximum signal speed c . No causal influence propagates faster.

Axiom 3: Coherence

Stable structure requires self-reinforcing, coherent propagation. Incoherent modes disperse. Among coherent states, the fundamental mode has minimal topological winding (Axiom 3b).

What the Axioms Derive

Result	Status	Key Fact
Three generations of matter	CONDITIONAL (0.85)	Algebra locks at $N=3$, but denominator theorem needs T2 bridge
Topological weights (2,1)	PARTIAL (0.85)	Closure order theorem proven, physical-realization bridge open
Koide phase $\delta \approx 2/9$	EMPIRICAL (0.65)	$ \delta - 2/9 = 7.4 \times 10^{-4}$ (0.003%). Strongest empirical anchor
Koide ratio $Q = 2/3$	DERIVED (0.95)	Geometric identity: 120° spacing forces $R/A = \sqrt{2}$
Gravity as refraction	DERIVED (0.95)	Randers/Finsler equivalence. Shapiro delay: 0.01% error
8-hour sleep constant	ARGUED (0.72)	$2/3$ active fraction plausible, but 8h not derived from axioms
Weinberg angle $\sin^2(\theta_W)$	DERIVED (0.90)	0.22310 via Axiom 3b. Matches PDG on-shell to 0.13σ
Fine structure constant α	ARGUED (0.35)	Casimir combination hits $1/137.119$ (0.061% error)
Propagation Lagrangian	CONDITIONAL (0.72)	Scalar-tensor EFT maps to Brans-Dicke in linear limit
Variable c prediction	ARGUED (0.65)	$c_{\text{local}} = 1/\sqrt{1+\lambda\chi}$. Constrained by Cassini to $\lambda \leq 10^{-24} \text{ m}^{-1}$
QCD confinement from λ_{QCD}	ARGUED (0.72)	Argued RG bridge from λ_{QCD} . 1-loop overshoots: 2.2 fm vs ~ 0.9 fm

The God Equation (CONDITIONAL, 0.88)

$$\lambda_{\text{QCD}} = \sqrt{2} \cdot l_P \cdot \exp(4\pi^2 \cdot N^{D/2} / b_0)$$

Predicts the top quark Compton wavelength from the Planck length alone. $N=3$, $D=3$, $b_0=16/3$. Predicted: 1.145×10^{-18} m. Measured: 1.14×10^{-18} m. Error: 0.4%. Zero free parameters. Key gap: H_{prod} (statistical independence) not yet proved. Path A: chiral projection needs Fourier-to-position-space bridge. Path B: actual closure object shows no near-decoupling.

How to Kill It

Find a 4th generation particle at any mass, any coupling. The framework says it cannot exist - the exclusion is topological, not kinematic.

Show Koide $Q \neq 2/3$ for neutrinos. Already falsified: $Q_{\text{NO}} = 0.55$, $Q_{\text{IO}} = 0.48$ (>5% deviation). Koide is EM phenomenon.

EEG phase transitions: If Critical Slowing Down does not precede cognitive insight in $\geq 7/10$ sessions, the cross-scale claim fails.

Tau g-2 at Belle II: Framework predicts a specific torsion correction. Pure QED agreement to 10^{-5} falsifies it.

What Failed (Honesty)

The harmonic series mass claim failed (CV=0.94, noise). The ϕ^3 electron/up ratio is interesting but uncertainty-limited ($p=0.007$). The spin-pair selection strike produced three independent no-go results. All documented in `sandbox_results.md`. A framework that only publishes successes is not science.